Reconstruction-Based Hand–Eye Calibration Using Arbitrary Objects

Shiyu Xing, Fengshui Jing, and Min Tan

***Abstract*—This article introduces a flexible hand–eye calibration technique for a 3-D sensor from a reconstruction perspective, with no need for a specialized and accurate calibration rig. Our intention is to find the hand–eye relation that simultaneously aligns multiview point clouds of a common scene into the robot base frame, namely simultaneous calibration and reconstruction. To achieve this goal, a novel variant of iterative closest point (ICP) algorithm based on the Gauss–Newton method and Lie algebra is proposed, which iteratively transforms multiview point clouds into the robot base frame, estimates point-to-point correspondences between point clouds then refines the hand–eye relation to minimize the Euclidean distance between corresponding points. In addition to the calibration result, it returns a preliminary reconstruction as a byproduct. Cases of degeneracy and applicable conditions are given and proved. Using arbitrary daily objects with no prior information and a real robotic eye-in-hand system, we verify our method feasible and effective.**

***Index Terms*—3-D reconstruction, arbitrary objects, computer vision and its industrial application, hand–eye calibration, iterative closest point (ICP) variant.**

I. Introduction

THE recent developments in low-cost 3-D sensors open up a whole new dimension for robotics and bring great potential for industrial automation. The hand–eye system consisting of an industrial robot and a 3-D sensor is a typical use case and is now widely deployed to carry out various vision-guided applications, e.g., robotic grasping [1], spray painting [2], and workpiece reconstruction [3]. To achieve fusion of 3-D measurements taken at different viewpoints into the robot base frame, the relation between sensor and robot hand needs to be accurately recovered, which is commonly known as hand–eye calibration.

Manuscript received 27 April 2022; revised 10 August 2022; accepted 28 August 2022. Date of publication 2 September 2022; date of current version 4 May 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62173327 and Grant U1813208, and in part by the National Key research and development program of China under Grant 2019YFB1312703. Paper no. TII-22-1805. *(Corresponding author: Fengshui Jing.)*

The authors are with the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing 100049, China, and also with the State Key Laboratory of Management and Control of Complex Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China (e-mail: xingshiyu2020@ia.ac.cn; fengshui.jing@ia.ac.cn; min.tan@ia.ac.cn).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TII.2022.3203771.

Digital Object Identifier 10.1109/TII.2022.3203771

Although the sensor is rigidly mounted at the end of the serial manipulator, the hand–eye relation does change over time and on-site calibration needs to be conducted in some occasions. The related works rely on a specialized calibration rig with high manufacturing accuracy or other special geometrical constraints and usually introduce certain operational complexity. As a result, the process of hand–eye calibration and reconstruction are normally separated, in the way calibration is conducted first as a basis for the following reconstruction task. Therefore, on one hand, we wish to propose an operationally friendly method without strict requirements on calibration targets, which makes on-site calibration easy. On the other hand, we wish to reformulate the calibration problem from a reconstruction perspective and propose a new iterative solution with a more intuitive geometrical meaning. The contributions of this article are threefold.

1) Reformulation of a hand–eye calibration problem for the 3-D sensor from a reconstruction perspective and a novel iterative closest point (ICP) variant as the corresponding solution, which does not rely on an accurate calibration rig with known dimensions. The proposed algorithm aligns multiple point clouds into the robot base frame by iterative refinement of the hand–eye relation. It returns the calibrated hand–eye relation together with a preliminary reconstruction as byproduct.
2) Introduction of the cases of degeneracy and applicable condition for the ICP variant based on mathematical reasonings.
3) Extensive experiments using arbitrary daily objects and real eye-in-hand system. The proposed method is proved feasible and effective.

II. Related Work

Hand–eye calibration has been extensively studied over decades since 1980 s, beginning with the well-known kinematic loop in form of $AX = XB$ first proposed by Shiu and Ahmad [4], where X denotes hand–eye transformation to be estimated, A stands for the relative motion between any two different robot poses A_i and A_j. B stands for the sensor motion, as is shown in Fig. 1.

In the initial stage, researchers decoupled the rotation from translation and applied linear algebra to estimate rotation and translation separately using rotation axis and angle [5], unit quaternions [6], and Lie Theory [7]. These research studies all claim that hand–eye transformation could be uniquely determined using at least two robot motions with unparallel rotation

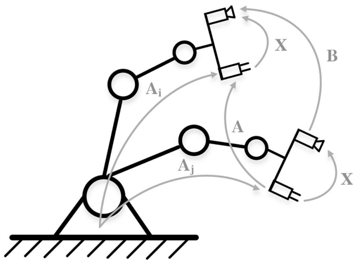


Fig. 1. Classic hand–eye calibration.

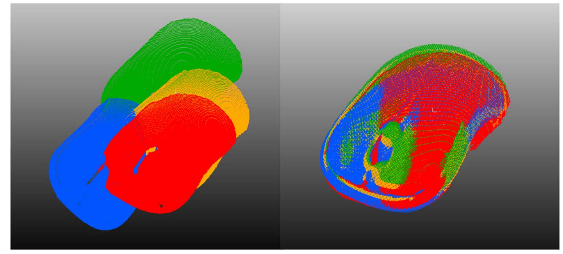


Fig. 2. Reconstruction of a mouse under the robot base frame, with incorrect hand–eye relation (left) and correct hand–eye relation (right).

axes. Later, to avoid error propagation from rotation estimation to translation, simultaneous solutions using dual quaternion [8] and Kronecker product [9] were presented. Further contributions focus on iterative optimization based on closed-form solution to improve calibration accuracy [10], [11], [12], [13], [14]. Although the experiments in above-mentioned studies are camera-based, the formulation of calibration problem in form of $AX = XB$ is essentially the rigid body motion, which is independent of sensor types and their working principle, thus they serve as a kind of universal solution and fail to give more geometrically meaningful objective function specific to sensor type during optimization, e.g., reprojection error for camera and Euclidean distance for 3-D sensors, which are more likely to provide higher calibration accuracy [15].

More specifically for 3-D sensors, an accurate calibration rig with known dimensions is usually a must in most of the relevant literature. Some typical instances are standard sphere [16], [17], calibration board [18] and free-form surface with design CAD model [19], [20]. High-precision machining can be costly. Besides, the specimen unavailability and accuracy degeneration render these methods impractical in some occasions. Other researchers utilized common geometrical constraints like plane [21], [22] and its unit normal vector to avoid independence on an accurate calibration rig. Nonetheless, at least three points on the plane need to be measured beforehand using an additional tool, which brings additional operation complexity. We noticed that an attempt given by Xu [23] uses the image-based scene features and achieves hand–eye calibration without a specialized calibration rig, but it fails to cope with featureless scenario and is not applicable for 3-D sensors without image output.

III. Methodology

A. Motivation

Normally the hand–eye calibration is carried out separately as a prerequisite for the following reconstruction task [3], [18], [24], [25] due to the reliance on a specialized calibration rig, but we find this separation unnecessary if one can achieve calibration by 3-D reconstruction of arbitrary objects. Our idea is straightforward, based on the fact that point clouds captured at different viewpoints should align, after transformation to robot frame using correct hand–eye relation and corresponding robot poses. Fig. 2 shows the reconstruction result of a stationary mouse under a robot frame in the case of correct calibration and incorrect calibration. Furthermore, as many industrial applications with eye-in-hand systems usually transform the measurements to robot frame before subsequent operations, we reformulate the calibration problem of 3-D sensors as finding the hand–eye relation, which simultaneously registers multiple point clouds of a common scene captured from different viewpoints, into the robot base frame. We seek to transfer the existing pattern, which is calibration for reconstruction, into reconstruction for calibration, achieving both processes simultaneously with calibration as the main purpose and preliminary reconstruction as a byproduct.

B. New ICP Variant

To achieve this goal, we proposed a novel ICP variant which is illustrated by Fig. 3 (in the case of two robot motions with three point clouds captured). Consider a serial manipulator with a rigidly mounted 3-D sensors making a series of n motions from the first robot pose to $(n+1)$th robot pose. A measurement in form of point cloud of a stationary object is taken at each robot poses. All the captured point clouds are transformed to the robot base frame by left multiplying the first initial guess of hand–eye relation $\boldsymbol{X}$ then corresponding robot poses $\boldsymbol{A}_i$, where both $\boldsymbol{X}$ and $\boldsymbol{A}_i$ are homogeneous transformation matrices in the special Euclidean group SE(3).

Then point-to-point correspondences between point clouds obtained before and after each robot motion are estimated. Specifically, find the closest point in one point cloud for each point in the other point cloud then update the point matches by discarding outliers with distance over a certain threshold.

The hand–eye relation is then refined to minimize the sum of squared-error on the distance between the coordinates of the estimated correspondences. In standard form

$$\begin{aligned} \min \quad & f(\boldsymbol{X}) = \sum_{i=1}^{n} \sum_{j=1}^{m_i} \|g_{ij}(\boldsymbol{X})\|^2 \\ & = \sum_{i=1}^{n} \sum_{j=1}^{m_i} \left\| \begin{bmatrix} \boldsymbol{R}_{A_i} & \boldsymbol{t}_{A_i} \end{bmatrix} \boldsymbol{X} \begin{bmatrix} \boldsymbol{p}_j^i \\ 1 \end{bmatrix} - \begin{bmatrix} \boldsymbol{R}_{A_{i+1}} & \boldsymbol{t}_{A_{i+1}} \end{bmatrix} \boldsymbol{X} \begin{bmatrix} \boldsymbol{q}_j^i \\ 1 \end{bmatrix} \right\|^2 \\ \text{s.t.} \quad & \boldsymbol{X} \in \mathrm{SE}(3) \end{aligned} \tag{1}$$

where n denotes the number of robot motions and m_i represents the number of correspondences between point clouds obtained before and after ith robot motion. $(\boldsymbol{p}_j^i, \boldsymbol{q}_j^i)$ stands for the jth correspondence in the m_i, where $\boldsymbol{p}_j^i$ is a point in point cloud obtained at ith robot pose (before ith motion) w.r.t. the sensor

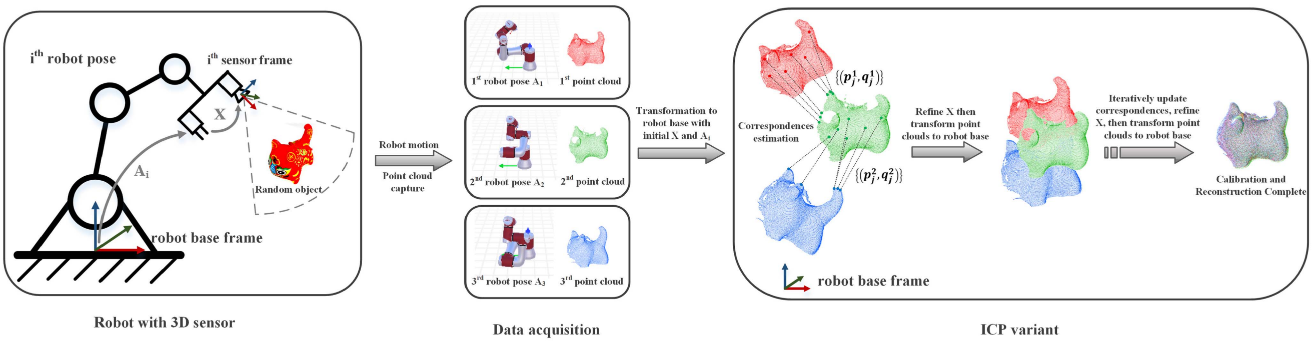


Fig. 3. Proposed hand–eye calibration method using the novel ICP variant, in case of three robot poses. Iteratively, correspondences between point clouds are estimated after transformation to the robot base frame, then hand–eye relation is refined using (1). It returns calibrated hand–eye relation and a preliminary reconstruction simultaneously.

frame and $\boldsymbol{q}_j^i$ is the corresponding point in point cloud obtained at $(i+1)$th robot pose (after ith motion) w.r.t. the sensor frame. $\boldsymbol{R}$ and $\boldsymbol{t}$ are the 3×3 rotation matrix in SO(3) and 3×1 translation vector respectively of the corresponding homogeneous transformation matrix (indicated by the subscript).

To solve this least square problem, the Gauss–Newton method is applied to optimize the rotation and translation iteratively and simultaneously. Rewrite the component function of (1) into form of rotation matrix and translation vector

$$\begin{aligned}
&\boldsymbol{g}_{ij}(\boldsymbol{R}_X,\boldsymbol{t}_X)\\
&=\begin{bmatrix}\boldsymbol{R}_{A_i} & \boldsymbol{t}_{A_i}\end{bmatrix}\begin{bmatrix}\boldsymbol{R}_X & \boldsymbol{t}_X\\ \boldsymbol{0} & 1\end{bmatrix}\begin{bmatrix}\boldsymbol{p}_j^i\\ 1\end{bmatrix}-\begin{bmatrix}\boldsymbol{R}_{A_{i+1}} & \boldsymbol{t}_{A_{i+1}}\end{bmatrix}\begin{bmatrix}\boldsymbol{R}_X & \boldsymbol{t}_X\\ \boldsymbol{0} & 1\end{bmatrix}\begin{bmatrix}\boldsymbol{q}_j^i\\ 1\end{bmatrix}\\
&=\boldsymbol{R}_{A_i}\boldsymbol{R}_X\boldsymbol{p}_j^i+\boldsymbol{R}_{A_i}\boldsymbol{t}_X+\boldsymbol{t}_{A_i}-\boldsymbol{R}_{A_{i+1}}\boldsymbol{R}_X\boldsymbol{q}_j^i-\boldsymbol{R}_{A_{i+1}}\boldsymbol{t}_X-\boldsymbol{t}_{A_{i+1}}.
\end{aligned} \tag{2}$$

As SO(3) is defined on multiplication rather than addition, we apply Lie Algebra so(3) and perturbation model to derive the Jacobian Matrix

$$\boldsymbol{J}_{ij}(\boldsymbol{\varphi}_X,\boldsymbol{t}_X)=\begin{bmatrix}-\boldsymbol{R}_{A_i}(\boldsymbol{R}_X\boldsymbol{p}_j^i)^{\wedge}+\boldsymbol{R}_{A_{i+1}}(\boldsymbol{R}_X\boldsymbol{q}_j^i)^{\wedge} & \boldsymbol{R}_{A_i}-\boldsymbol{R}_{A_{i+1}}\end{bmatrix} \tag{3}$$

where $\boldsymbol{\varphi}_X$ is the left perturbation on $\boldsymbol{R}_X$ and $(\cdot)^{\wedge}$ is the skew-symmetric matrix. Stack all the component functions and corresponding Jacobian matrices, we have

$$\boldsymbol{g}=\begin{bmatrix}\boldsymbol{g}_{11}^{\mathrm{T}}, & \boldsymbol{g}_{12}^{\mathrm{T}},\ldots, & \boldsymbol{g}_{1m_1}^{\mathrm{T}}, & \boldsymbol{g}_{21}^{\mathrm{T}},\ldots,\boldsymbol{g}_{nm_n}^{\mathrm{T}}\end{bmatrix}^{\mathrm{T}}\Bigg|_{\left(3\sum_{i=1}^{n}m_i\right)\times 1} \tag{4}$$

$$\boldsymbol{J}=\begin{bmatrix}\boldsymbol{J}_{11}^{\mathrm{T}}, & \boldsymbol{J}_{12}^{\mathrm{T}},\ldots, & \boldsymbol{J}_{1m_1}^{\mathrm{T}}, & \boldsymbol{J}_{21}^{\mathrm{T}},\ldots,\boldsymbol{J}_{nm_n}^{\mathrm{T}}\end{bmatrix}^{\mathrm{T}}\Bigg|_{\left(3\sum_{i=1}^{n}m_i\right)\times 6}. \tag{5}$$

Then, the normal equation reads

$$\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J}\Delta\boldsymbol{X}=-\boldsymbol{J}^{\mathrm{T}}\boldsymbol{g} \tag{6}$$

where

$$\Delta\boldsymbol{X}=\begin{bmatrix}\Delta\boldsymbol{\varphi}^{\mathrm{T}} & \Delta\boldsymbol{t}^{\mathrm{T}}\end{bmatrix}^{\mathrm{T}} \tag{7}$$

in which $\Delta\boldsymbol{\varphi}$ is the update perturbation in so(3) for rotation and $\Delta\boldsymbol{t}$ is the translation update. The $\Delta\boldsymbol{X}$ can be obtained as

$$\Delta\boldsymbol{X}=-\left(\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J}\right)^{-1}\boldsymbol{J}^{\mathrm{T}}\boldsymbol{g} \tag{8}$$

and is subsequently used to update $\boldsymbol{R}_X$ and $\boldsymbol{t}_X$. Then, go to (2) and repeat until $\Delta\boldsymbol{X}$ is small enough. The refined hand–eye relation $\boldsymbol{X}$ is utilized to transform the point clouds to the robot base frame for correspondences estimation in the next iteration.

The process of point clouds transformation to the robot base frame, correspondences estimation, and hand–eye relation refinement to minimize (1) are carried out in sequence iteratively until the average point-to-point distance of all the correspondences between two iterations is close enough. Although the correspondences estimation is generally incorrect at initial stage, it refines the hand–eye relation, which, in return, refines the correspondences estimation in the next iteration.

This ICP variant is summarized in algorithm1, where the initial guess can be roughly obtained via the dimensions of sensor holding flange and of sensor provided by the vendor. The $\exp(\cdot)$ denote the exponential mapping [26] from so(3) to SO(3). Practically, at least two robot motions with unparallel rotation axes are required for calibration purpose when using this ICP variant and the proof is given in the later section.

The proposed algorithm is similar to classic ICP [27] in the way that both algorithm update correspondences then solve the optimization problem iteratively, but they differ in details. The classic ICP aligns two point clouds, under the coordinate frame of the reference point cloud, by iteratively refining the point clouds transformation (can be sensor motion, etc.) to be estimated, and returns the best estimation of point cloud transformation when registration is complete. The proposed ICP variant aligns multiple point clouds simultaneously (no less than 3 for calibration purpose), under the robot base frame, by iteratively refining hand–eye relation, and returns the best estimation of hand–eye relation when simultaneous registration of multiple point clouds is complete.

It is worth noticing that although the proposed algorithm still converges with only two point clouds if we apply pseudoinverse instead of inverse to compute the update $\Delta\boldsymbol{X}$, the obtained $\boldsymbol{X}$ after convergence can not be taken as calibrated hand–eye

Algorithm 1: The Proposed ICP Variant.

Require: The $n+1$ robot poses during n robot motions $\{\boldsymbol{A_1}, \boldsymbol{A_2}, \ldots, \boldsymbol{A_{n+1}}\}$ and corresponding point clouds captured at each poses.
Require: Initial guess $(\boldsymbol{R}_X, \boldsymbol{t}_X)$
Require: Stop threshold ϵ_1 and ϵ_2
Require: Initialized $ratio = 0$, $counter = 0$, $||\Delta \boldsymbol{X}|| = +\infty$
while $ratio \leq \epsilon_1$ **do**
  Transform the point clouds to the robot base frame using $(\boldsymbol{R}_X, \boldsymbol{t}_X)$ and robot poses. Then update all the point-to-point correspondence in each motion $\{(\boldsymbol{p}_j^i, \boldsymbol{q}_j^i)\}, i = 1, 2, \ldots, n (n \geq 1), j = 1, 2, \ldots, m_i$
  while $||\Delta \boldsymbol{X}|| \geq \epsilon_2$ **do**
    $\boldsymbol{g}_{ij} \longleftarrow \boldsymbol{A}_i, \boldsymbol{A}_{i+1}, (\boldsymbol{p}_j^i, \boldsymbol{q}_j^i), \boldsymbol{R}_X, \boldsymbol{t}_X$
    $\boldsymbol{g} \longleftarrow \boldsymbol{g}_{11}, \boldsymbol{g}_{12}, \ldots, \boldsymbol{g}_{1m_1}, \boldsymbol{g}_{21}, \ldots, \boldsymbol{g}_{nm_n}$
    $\boldsymbol{J}_{ij} \longleftarrow \boldsymbol{A}_i, \boldsymbol{A}_{i+1}, (\boldsymbol{p}_j^i, \boldsymbol{q}_j^i), \boldsymbol{R}_X$
    $\boldsymbol{J} \longleftarrow \boldsymbol{J}_{11}, \boldsymbol{J}_{12}, \ldots, \boldsymbol{J}_{1m_1}, \boldsymbol{J}_{21}, \ldots, \boldsymbol{J}_{nm_n}$
    if $n = 1$ **then**
      $\Delta \boldsymbol{X} = -(\boldsymbol{J}^{\mathrm{T}} \boldsymbol{J})^{+} \boldsymbol{J}^{\mathrm{T}} \boldsymbol{g}$
    else
      $\Delta \boldsymbol{X} = -(\boldsymbol{J}^{\mathrm{T}} \boldsymbol{J})^{-1} \boldsymbol{J}^{\mathrm{T}} \boldsymbol{g}$
    end if
    Update:
    $\boldsymbol{R}_X = exp(\Delta \boldsymbol{\varphi}^{\wedge}) \boldsymbol{R}_X$
    $\boldsymbol{t}_X = \boldsymbol{t}_X + \Delta \boldsymbol{t}$
  end while
  $counter = counter + 1$
  $avg \longleftarrow$ Average of the distance between corresponding points
  if $counter = 1$ **then**
    $avgLst = avg$
  else
    $ratio = avg / avgLst$
    $avgLst = avg$
  end if
end while
return $\boldsymbol{R}_X, \boldsymbol{t}_X$

relation. The reason will be given in the next section and relevant experiment will be presented in Section IV.

C. Cases of Degeneracy and Applicable Condition

In this section, we first introduce relevant lemma and proposition derived from our observations, then cases of degeneracy and applicable conditions are presented.

Lemma 1: For any two unidentical rotations $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ in SO(3)

$$\mathrm{rank}(\boldsymbol{R}_1 - \boldsymbol{R}_2) = 2 \tag{9}$$

Proof: From algebraic perspective:

$$\mathrm{rank}(\boldsymbol{R}_1 - \boldsymbol{R}_2) = \mathrm{rank}\left(\boldsymbol{R}_1(\boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2)\right). \tag{10}$$

As the product of two matrices is the linear combination of column or row vectors of the factor matrices, the rank of the product is no higher than either of the factor matrices. Besides,

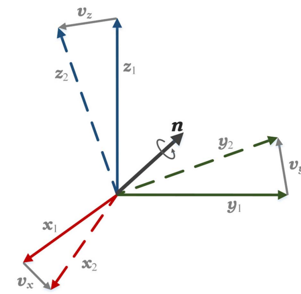


Fig. 4. Geometrical relation between two random rotations.

$\boldsymbol{R}_1^{-1}\boldsymbol{R}_2$ is a nonidentity rotation matrix with eigenvalue 1, we have

$$\mathrm{rank}(\boldsymbol{R}_1 - \boldsymbol{R}_2) \leq \min\left(\mathrm{rank}(\boldsymbol{R}_1), \mathrm{rank}(\boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2)\right) = 2. \tag{11}$$

In the meanwhile

$$\begin{aligned}
\mathrm{rank}(\boldsymbol{R}_1 - \boldsymbol{R}_2) &= \mathrm{rank}\begin{bmatrix} \boldsymbol{0} & \boldsymbol{I}_3 \\ -\boldsymbol{R}_1(\boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2) & \boldsymbol{0} \end{bmatrix} - 3 \\
&= \mathrm{rank}\begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{I}_3 \\ -\boldsymbol{R}_1(\boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2) & \boldsymbol{0} \end{bmatrix} - 3 \\
&= \mathrm{rank}\begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{I}_3 \\ \boldsymbol{0} & \boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2 \end{bmatrix} - 3 \\
&\geq \mathrm{rank}\begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2 \end{bmatrix} - 3 \\
&= \mathrm{rank}(\boldsymbol{R}_1) + \mathrm{rank}(\boldsymbol{I} - \boldsymbol{R}_1^{-1}\boldsymbol{R}_2) - 3 \\
&= 2.
\end{aligned} \tag{12}$$

Combined with foregoing (11), Lemma 1 is derived.

From a geometrical perspective, rewrite $\boldsymbol{R}_1$ and $\boldsymbol{R}_1$ into

$$\begin{aligned}
\boldsymbol{R}_1 &= \begin{bmatrix} \boldsymbol{x}_1, & \boldsymbol{y}_1, & \boldsymbol{z}_1 \end{bmatrix} \\
\boldsymbol{R}_2 &= \begin{bmatrix} \boldsymbol{x}_2, & \boldsymbol{y}_2, & \boldsymbol{z}_2 \end{bmatrix}
\end{aligned} \tag{13}$$

where $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$ are the direction cosine of three axes, respectively. Subsequently,

$$\boldsymbol{R}_1 - \boldsymbol{R}_2 = \begin{bmatrix} \boldsymbol{x}_1 - \boldsymbol{x}_2, & \boldsymbol{y}_1 - \boldsymbol{y}_2, & \boldsymbol{z}_1 - \boldsymbol{z}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{v}_x, & \boldsymbol{v}_y, & \boldsymbol{v}_z \end{bmatrix}. \tag{14}$$

As is shown in Fig. 4, any two rotations can be converted to each other via a rotation through some angle about an axis $\boldsymbol{n}$, thus $\boldsymbol{v}_x$, $\boldsymbol{v}_y$, and $\boldsymbol{v}_z$ are all orthogonal to $\boldsymbol{n}$, implying they are coplanar and of rank 2.

Proposition 1: For a set of n rotations $(\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_n)$ in which any two can be converted to each other by rotation through some angle about a common axis. The following equation always

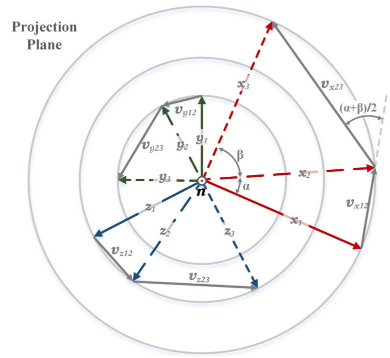


Fig. 5. Projection along the common rotation axis. A degenerative case.

holds:

$$\operatorname{rank}\begin{bmatrix} \boldsymbol{R}_1-\boldsymbol{R}_2 \\ \boldsymbol{R}_1-\boldsymbol{R}_3 \\ \cdots \\ \boldsymbol{R}_1-\boldsymbol{R}_n \\ \boldsymbol{R}_2-\boldsymbol{R}_3 \\ \boldsymbol{R}_2-\boldsymbol{R}_4 \\ \cdots \\ \boldsymbol{R}_{n-1}-\boldsymbol{R}_n \end{bmatrix}=2. \tag{15}$$

Proof: Without loss of generality, we present geometrical proof using three rotations meeting the condition above. Following the denotation used in the geometrical proof of Lemma 1, Fig. 5 shows the projection view along the common rotation axis $\boldsymbol{n}$, about which three rotations can be converted to each other by rotation.

Assume the rotation angle from $\boldsymbol{R}_1$ to $\boldsymbol{R}_2$ is α and the rotation angle from $\boldsymbol{R}_2$ to $\boldsymbol{R}_3$ is β. According to lemma 1, $\boldsymbol{v}_{x12}$, $\boldsymbol{v}_{x23}$, $\boldsymbol{v}_{y12}$, $\boldsymbol{v}_{y23}$, $\boldsymbol{v}_{z12}$ and $\boldsymbol{v}_{z23}$ all lie on the projection plane and one must have

$$\lambda_1\boldsymbol{v}_{x12}+\lambda_2\boldsymbol{v}_{y12}+\lambda_3\boldsymbol{v}_{z12}=\boldsymbol{0} \tag{16}$$

where λ_1, λ_2, and λ_3 are a set of scalars, which are not 0 simultaneously. Obviously, rotate $\boldsymbol{v}_{x12}$ by $(\alpha+\beta)/2$ then make an extension by a constant scale factor $\frac{\sin(\beta/2)}{\sin(\alpha/2)}$, we have $\boldsymbol{v}_{x23}$

$$\boldsymbol{v}_{x23}=\frac{\sin(\beta/2)}{\sin(\alpha/2)}\begin{bmatrix}\cos((\alpha+\beta)/2) & -\sin((\alpha+\beta)/2)\\ \sin((\alpha+\beta)/2) & \cos((\alpha+\beta)/2)\end{bmatrix}\boldsymbol{v}_{x12}. \tag{17}$$

Same relation applies to $\boldsymbol{v}_{y12}$ and $\boldsymbol{v}_{y23}$, $\boldsymbol{v}_{z12}$ and $\boldsymbol{v}_{z23}$ as well. As a result

$$\lambda_1\boldsymbol{v}_{x23}+\lambda_2\boldsymbol{v}_{y23}+\lambda_3\boldsymbol{v}_{z23}=\boldsymbol{0}. \tag{18}$$

That is to say, for a set of rotations in which anyone can be converted to the others by rotation about a common axis, the column vectors in (15) share the same linear correlation with the column vectors in $(\boldsymbol{R}_i-\boldsymbol{R}_j)$, where $\boldsymbol{R}_i$ and $\boldsymbol{R}_j$ are two arbitrary rotations in the set. Thus, the rank of (15) is 2.

Alternatively, one can see that the transposes of the *n* rotation matrices also meet the condition in Proposition 1, after some manipulation. This implies the row space of (15) is a plane of rank 2. ■

According to the above lemma and proposition, as the Jacobian matrix in (5) is the stack of (3), it will not be full column rank if the robot makes only one motion (two point clouds captured at two different robot poses) or several motions about common rotation axis or pure translation. In these degenerative occasions, the coefficient matrix $\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J}$ will be singular with an infinite condition number, which means the obtained update using (8) can be unstable making the algorithm hardly converge. Although the pseudoinverse is able to stabilize the update and render the convergence possible, the finally returned $\boldsymbol{X}$ is not the correct hand–eye relation. According to the previous hand–eye calibration studies based on rigid body motion [4], [5], [6], [7], infinite solutions exist in the above scenarios, which implies the returned $\boldsymbol{X}$ is only one of the infinite global optimums that is able to align multiple point clouds after transformation to the robot base frame, in the degenerative cases.

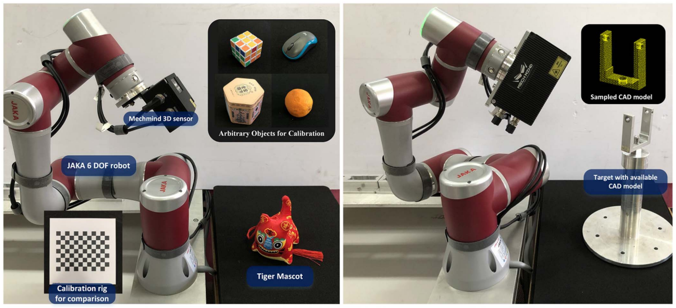
Besides, one can see that when motions are not about common rotation axis, the row space of (15) is seen to be multiple different planes (of rank 3). Jointly considering the aforementioned cases of degeneracy, at least two robot motions with unparallel rotation axes are requisite, which is the applicable condition of the proposed ICP variant for hand–eye calibration purposes. Practically, it is not likely to encounter these degenerative situations especially when multiple robot poses, thus few challenges are posed to the practicality of the proposed method.

IV. Experiments

A. System Setup and Hand–Eye Calibration

We verified the performance of the proposed method with a robotic eye-in-hand system consisting of a 6 DOF serial manipulator and a snapshot 3-D sensor. To demonstrate our method's independence on accurately manufactured calibration rig and any other prior knowledge of them, some arbitrary daily objects with no prior information were used during the calibration process, including an orange, a Chinese mascot of tiger year, a snack box, a Rubik's cube, and a wireless mouse, as is shown in Fig. 6(a). Two-dimensional calibration rig-like checkerboard [18], [24], [28] and fiducial marker [1], [19] are commonly used in 3-D sensor hand–eye calibration. In order to present a more intuitive demonstration on the performance of the proposed method, we compared our method with the one recently proposed by Wu [28], which used an accurate calibration checkerboard and outperforms various classic and recent hand–eye calibration techniques. A similar 12×9 checkerboard with side length of 10 mm ($\pm$0.01mm) was used as accurate calibration rig to carry out the comparison experiment, shown in Fig. 6(a).

For each of the arbitrary objects, eight point clouds were captured at eight different robot poses and downsampled using a voxel grid filter with a leaf size of 0.001 m. The background plane was segmented out; then, the calibration was carried out using ICP variant and different numbers of processed point clouds,



(a) (b)

Fig. 6. System setup. (a) Setup for hand–eye calibration. (b) Pose measurement experiment for accuracy assessment.

from a minimum number of 2 to total 8, where the case of two point clouds is degenerative and pseudoinverse was applied to facilitate the convergence in that case.

The initial guess was set roughly as

$$\boldsymbol{X} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{19}$$

and the stop threshold $ratio$, which is the average of the distance between two points of all the correspondences at current iteration over that at last iteration, was set to 0.98.

The average of the distance between all the corresponding points over iteration, in the case of different number of point cloud used in calibration, is plotted in Fig. 7. One can see that the proposed ICP variant converged reliably in all cases. Even in the degenerative case where only two point clouds were utilized, the pseudoinverse successfully stabilized the update and made convergence possible.

To help visualize the process of simultaneous alignment of multiple point clouds, the point clouds transformed to the robot base frame at some of the iterations during ICP variant using all the eight point clouds (in different colors) are plotted in Fig. 8. The point clouds successfully aligned as the algorithm converged. The calibration result and preliminary reconstruction were obtained when the algorithm stopped.

The same robot poses were used for data collection in the comparison experiment with a calibration rig. All calibration results, including the proposed reconstruction-based ones and calibration rig-based ones, were evaluated by the pose measurement experiment in the next section.

B. Accuracy Assessment

It is widely accepted that the ground truth of hand–eye relation is hardly available; therefore, the calibration accuracy is usually evaluated from the reconstruction and application perspective [25], [29]. Inspired by these works, a pose measurement experiment is designed for accuracy evaluation. Point clouds of a stationary standard mechanical target with a known CAD model (see Fig. 6(b)) were captured at n random robot poses differently

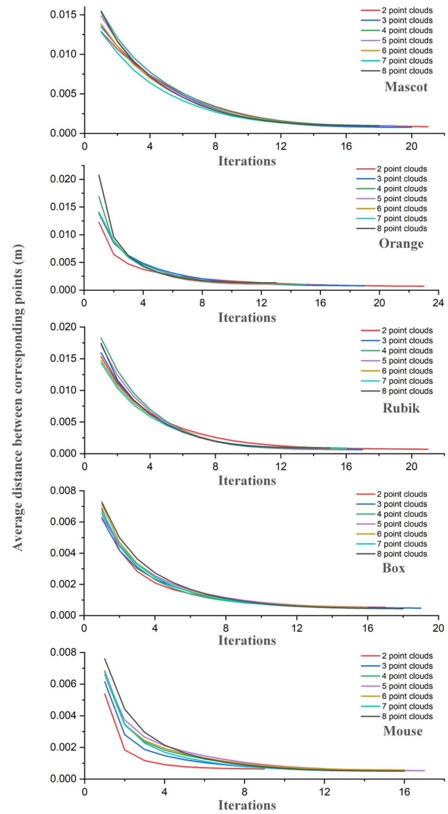


Fig. 7. Average of the distance between corresponding points over iterations during the calibration process. Each subplot contains eight curves in different colors indicating a different number of point clouds used in the calibration, from 2 to 8. The subplots from top to bottom show the calibration using mascot tiger year, orange, Rubik's cube, snack box, and wireless mouse in sequence.

than those utilized for calibration. Then, we transformed the n point clouds into the robot base frame using corresponding hand poses and calibration result to be assessed. Then pose of each 3-D scan, w.r.t. the robot base frame, was measured using the sampled CAD model via sample consensus initial alignment (SAC-IA) with fast point feature histograms (FPFH) features [30] then ICP registration.

Ideally, the n measurement results of the stationary target are consistent with correct hand–eye relation, thus the standard deviation (SD) of the measured rotations and positions are taken as the error metric, defined as

$$\mathrm{Err}_{\boldsymbol{R}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} ||\log(\boldsymbol{R}_i^{\mathrm{T}}\tilde{\boldsymbol{R}})||^2} \tag{20}$$

$$\mathrm{Err}_{\boldsymbol{t}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} ||\boldsymbol{t}_i - \tilde{\boldsymbol{t}}||^2} \tag{21}$$

where $\log(\cdot)$ defines the logarithm mapping [26] from SO(3) to so(3). $\boldsymbol{R}_i$ and $\boldsymbol{t}_i$ are the rotation component and position component of ith measurement respectively. The rotation and position averaging of all the n measured poses are denoted as $\tilde{\boldsymbol{R}}$ and $\tilde{\boldsymbol{t}}$.

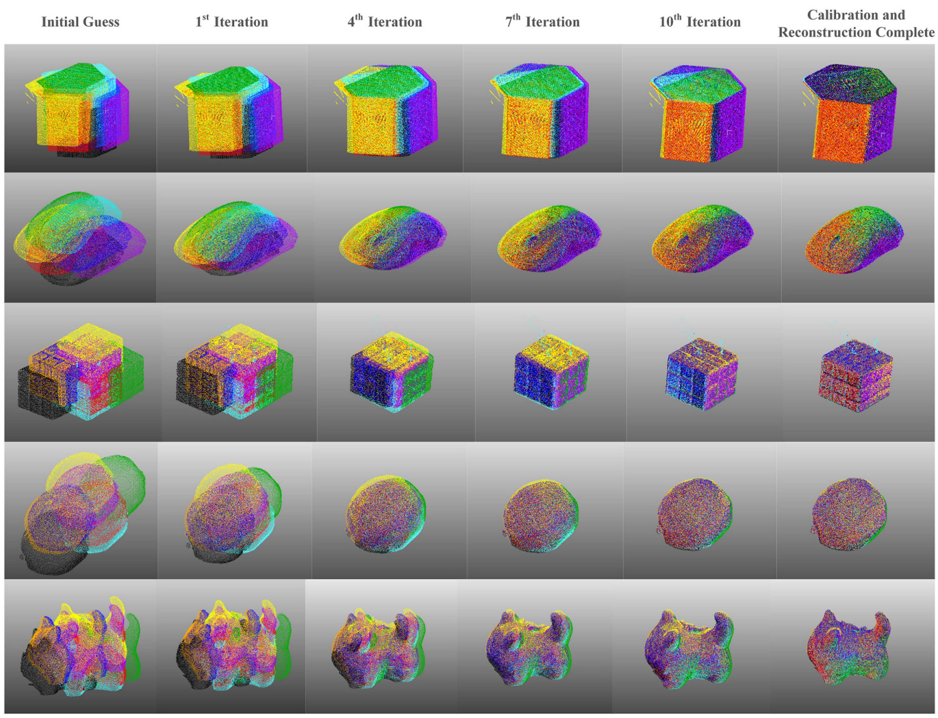


Fig. 8. Visualization of the alignment process during the calibration using the proposed ICP variant and all eight point clouds, at some iterations. From top row to bottom row, the objects are snack box, wireless mouse, Rubik's cube, orange, and mascot of tiger year in sequence. All point clouds are under the robot base frame.

TABLE I
ACCURACY ASSESSMENT USING TRANSLATION ERROR METRIC ERR$_t$ (UNIT:MM)

Objects	Number of point clouds/images							Uncalibrated
	2	3	4	5	6	7	8	
Box	88.5617	6.2298	3.9205	2.2485	2.0332	1.7917	1.7713	
Mascot	12.9308	2.7660	2.2397	2.0635	1.8255	1.7762	1.8420	
Mouse	35.6991	3.9322	2.1431	1.8307	1.8634	1.5730	1.5872	34.6116
Orange	23.5732	10.2851	5.1843	3.1182	2.0593	1.9213	1.6248	
Rubik	19.3936	4.6429	4.1239	2.3106	2.3923	2.5694	1.8641	
Calibration Rig	n/a	4.1491	2.3354	2.2300	2.1261	2.1141	2.2397	

TABLE II
ACCURACY ASSESSMENT USING ROTATION ERROR METRIC ERR$_R$ (UNIT:°)

Objects	Number of point clouds/images							Uncalibrated
	2	3	4	5	6	7	8	
Box	3.3270	3.0278	0.8544	0.3791	0.3486	0.3103	0.3220	
Mascot	2.3349	1.1124	1.04377	0.9154	0.7128	0.5863	0.5090	
Mouse	4.7751	1.4596	0.3339	0.3136	0.3246	0.3731	0.4088	4.0616
Orange	7.6814	4.7663	2.0980	1.1838	0.8089	0.7325	0.4482	
Rubik	6.1724	1.8790	1.7760	0.8516	0.6632	0.7305	0.4401	
Calibration Rig	n/a	0.5115	0.4978	0.3628	0.3491	0.3115	0.3113	

In this experiment, scans of the target were taken from 50 random viewpoints ($n = 50$). All the calibration results obtained were grouped according to the number of data used during calibration and further subgrouped based on the calibration object. For each of the calibration results, the pose measurement experiment was conducted, and the rotation and position deviation of all 50 measurements, from the average, are shown in a corresponding box plot in Fig. 9.

Err$_R$ and Err$_t$, namely the root mean square of 50 rotation and position deviations or the standard deviation of 50 measured rotations and positions, in the case of different calibration object and different number of point clouds/images utilized are presented in Tables I and II, respectively.

According to above calibration process and accuracy assessment results, we have the following discussion.

1) In general, the proposed algorithm converged successfully and aligned multiview point clouds. The accuracy of the proposed reconstruction-based method is inferior to the calibration rig-based method when a small number of calibration data were used, especially in rotation accuracy. One can also see that the proposed method shows comparable accuracy as the number of data used for calibration

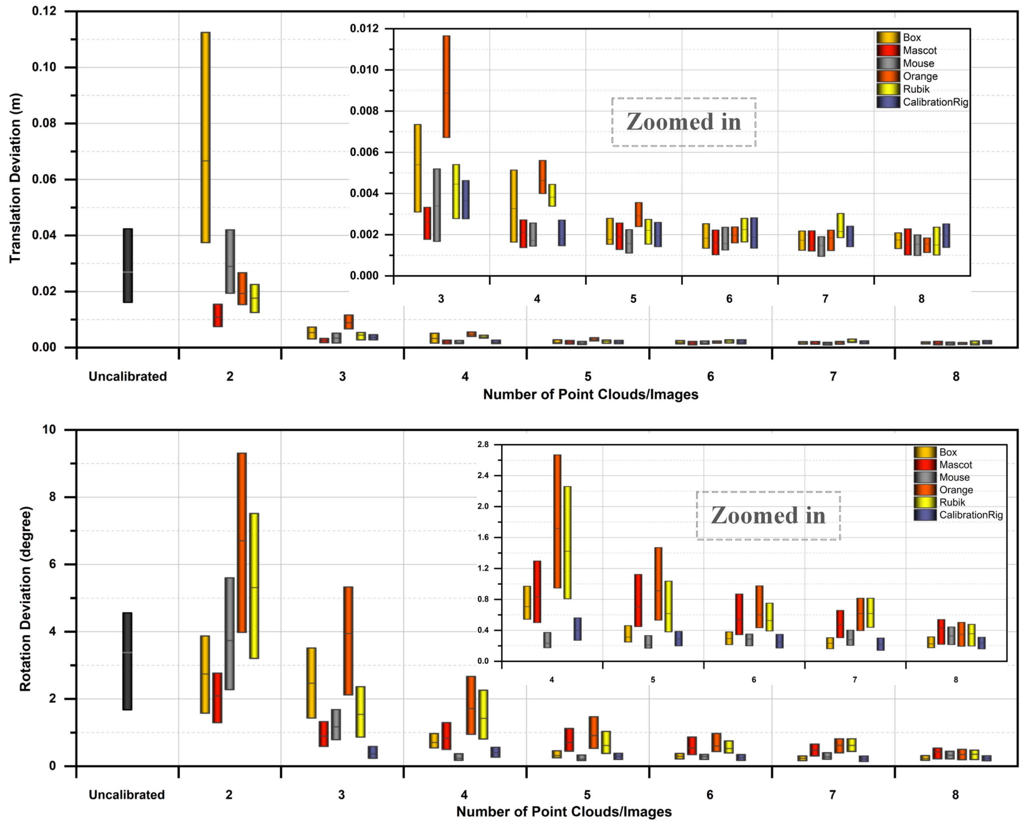


Fig. 9. Result of the pose measurement experiment in the case of different calibration results. The plot above is position-related and the one below is rotation relevant. Each box represents the distribution of 50 deviations, from the first quartile to the third quartile, with a median line in between. The uncalibrated group shows the pose measurement result using initial guess.

increases. The error when all eight point clouds were used is around 1.7 mm and 0.4° while the calibration rig-based method shows an error of 2.24 mm and 0.31°. The proposed algorithm works with arbitrary objects and the hand–eye relation is iteratively refined during the 3-D reconstruction.

2) It is worth mentioning that the standard mechanical target used in the above accuracy assessment is a mechanical part to be disassembled and reassembled by a robot in a national key research and development program we are working on. The proposed method is already applied in the project and supports the pose measurement and grasping operation reliably. We believe it is applicable for tasks without strict accuracy requirements, e.g., robotic grasping [31] and spray painting [32]. For robotic 3-D reconstruction, multiview point clouds of reconstruction targets are naturally needed and the reconstruction-based method is able to make calibration an online process of reconstruction, thus avoiding the unnecessary separation of two processes.

3) Although the pseudoinverse stabilized the update and made the algorithm converge successfully, in the degenerative two-point cloud case, the accuracy of obtained results can be worse than the uncalibrated initial guess in most occasions. As introduced in Section III, the returned result is only one of the infinite solutions that are able to align two captured point clouds under the robot base frame; therefore, the algorithm is not applicable for hand–eye calibration purpose when only two point clouds are captured. At least two robot motions with unparallel rotation axes (three point clouds captured) are recommended.

4) The accuracy of calibration using orange is lower than that using other objects in the case of a small number of point clouds. We assume that the algorithm tends to converge to local minimum as the orange is roughly a sphere without distinct 3-D features. As the number of point clouds increases, the accuracy becomes comparable with other objects, indicating the proposed algorithm is applicable without strict requirements on calibration object, when multiple point clouds are involved.

C. Runtime

Processing multiple point clouds simultaneously and iteratively can be computationally expensive, so we evaluated the complexity from a runtime perspective. All algorithms were coded with C++ and Point Cloud Library into a single-core implementation on a Dell Precision 7540 notebook featuring an Intel Core i7 9750H CPU and 16 GB of memory. The time consumption using different arbitrary objects and a different number of point clouds are given in Table III.

TABLE III
TIME CONSUMPTION USING ARBITRARY OBJECTS AND DIFFERENT NUMBER OF POINT CLOUDS

Object	Number of point clouds	Av. Corr./Iter.	Iter.	Runtime
Mascot	2	16 540	22	1.760 s
	3	34 114.9	20	3.313 s
	4	50 230.7	20	5.102 s
	5	66 369.1	19	6.434 s
	6	829 42	19	7.995 s
	7	100 427	18	9.254 s
	8	117 134	18	10.943 s
Box	2	23 060	11	1.210 s
	3	49 896.3	9	2.187 s
	4	73 831.7	9	3.391 s
	5	97 772.4	10	4.886 s
	6	121 921	10	6.067 s
	7	147 859	11	8.174 s
	8	173 363	11	9.498 s
Mouse	2	7777	8	0.329 s
	3	14 957	14	0.993 s
	4	22 134	15	1.526 s
	5	29 311	16	2.129 s
	6	36 649	15	2.539 s
	7	44 956	15	3.106 s
	8	53 050	16	4.013 s
Orange	2	7288.7	34	1.121 s
	3	14 339.7	17	1.158 s
	4	21 786.1	13	1.455 s
	5	29 333.3	13	1.949 s
	6	36 997.5	12	2.375 s
	7	44 266.5	13	2.957 s
	8	51 570.9	13	3.465 s
Rubik	2	11 698	22	1.318 s
	3	23 804.8	20	2.409 s
	4	34 592	17	3.181 s
	5	45 386	17	4.093 s
	6	57 724.1	15	4.803 s
	7	70 274.2	14	5.522 s
	8	82 254.3	15	7.191 s

In general, the algorithm stopped within 10 s and the runtime is positively related to the average number of total correspondences processed per iteration and number of iterations. The maximum time cost is 10.943 s when 8 mascot point clouds were used. The algorithm handled 117134 correspondences per iteration and stopped after 18 iterations.

D. Impact of Robot Positioning Performance on Accuracy

This section discusses the relation between robot's performance in rotation and translation and accuracy of proposed calibration. Artificial rotation error (up to 1° with an interval of 0.1° about a random axis) and translation error (up to 10 mm with an interval of 1 mm in a random direction) were added to all eight robot poses. For each of the error levels, we carried out 20 rounds of hand–eye calibration using the proposed method and all 8 mouse point clouds. Then all the calibration results went through the accuracy assessment. Accuracies of the 20 returned calibration results at different error levels are given in Fig. 10. Each box represents the error from the first quartile to the third quartile of 20 hand–eye relations calibrated with robot poses at a certain error level, with a median line in between. The caps show the minimum and maximum errors.

One can see that both $\mathrm{Err}_{\boldsymbol{R}}$ and $\mathrm{Err}_{\boldsymbol{t}}$ of the calibration result increase approximately linearly as robot translation error or rotation error increase, which shows certain dependence of the proposed method on robot positioning accuracy. For a robot

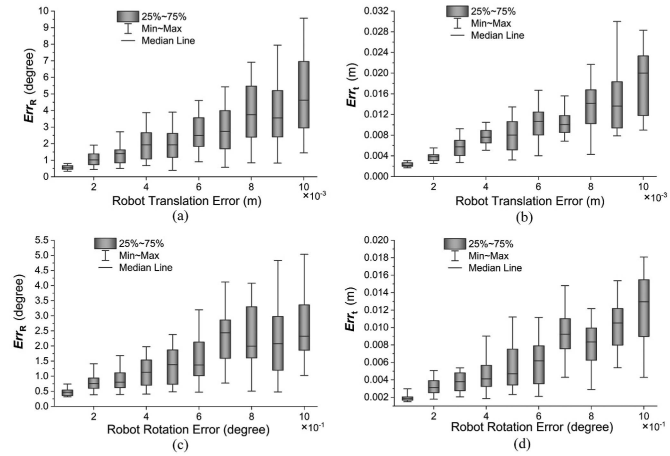


Fig. 10. Impact of robot positioning performance on calibration accuracy. (a) $\mathrm{Err}_{\boldsymbol{R}}$ against robot translation error. (b) $\mathrm{Err}_{\boldsymbol{t}}$ against robot translation error. (c) $\mathrm{Err}_{\boldsymbol{R}}$ against robot rotation error. (d) $\mathrm{Err}_{\boldsymbol{t}}$ against robot rotation error.

with low positioning accuracy, the proposed method may fail to calibrate the hand–eye relation accurately.

E. Tolerance to Initial Guess

Point cloud registration algorithms are usually sensitive to initial guess. Irrational initial guess may lead the algorithms to local optimum, ending with incorrect registration. This section tests the algorithm's tolerance to the initial guess. For each of the arbitrary calibration objects, 100 random initial values of hand–eye relation (with a random error of up to 50 mm for each of *x, y,* and *z* in position and a random error of up to 10° in rotation) were utilized to run the algorithm with all eight point clouds. The registration processes in the first 10 runs of the 100 with 8 mouse point clouds are shown in Fig. 11(a) for reference. The camera position is fixed using Point Cloud Library; thus, point clouds transformed to the robot base frame using random initial hand–eye relation appear randomly in the first screenshot of each row. According to observation, all 500 runs stopped with correct registration, and all returned hand–eye relations went through the accuracy assessment with the aforementioned error metric, results are shown in Fig. 11(b). The worst case presents an error of 2.53 mm and 0.58°.

It is worth noticing that the range setting of the initial value in this experiment is relatively rough compared with the dimensions of the sensor holding flange and is able to cover most of the cases especially when the CAD model is available. Spherical objects without distinct 3-D features like orange can be quite challenging for classic ICP, whereas it poses little challenges to the proposed ICP variant. All 100 runs with random initial hand–eye relation and all 8 orange point clouds returned successful registration and accurate calibration result.

V. CONCLUSION

We introduced a hand–eye calibration method for 3-D sensors, which does not rely on any specialized calibration rig but uses the arbitrary daily object. The core of our method is a novel variant of the ICP algorithm, which achieves simultaneous registration

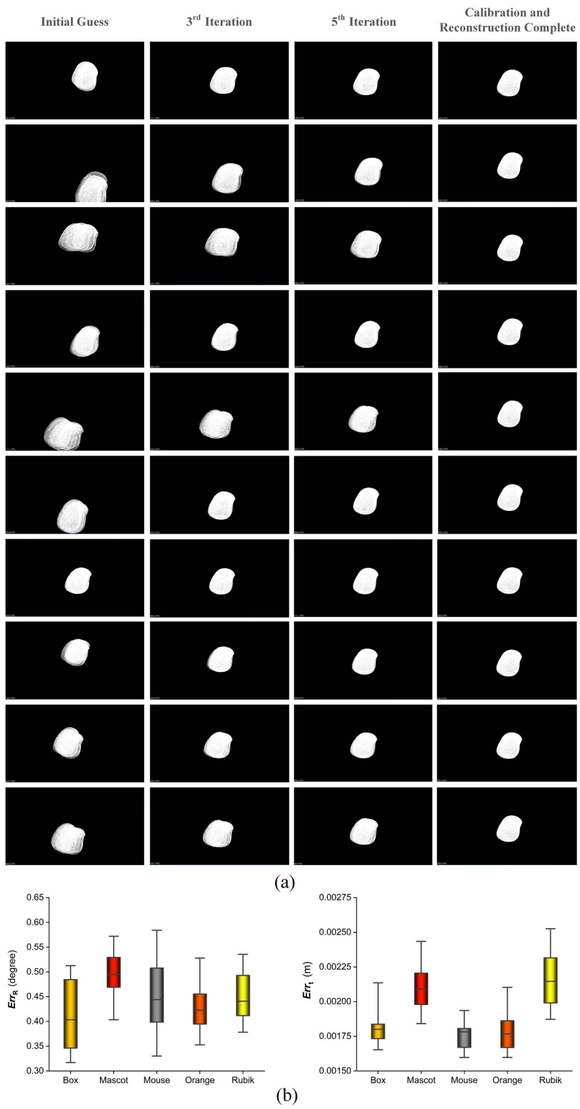


Fig. 11. (a) First 10 runs of the 100, using random initial hand–eye relation and 8 mouse point clouds. Each row shows screenshots of eight point clouds transformed to the robot base frame initially, after the third iteration, after the fifth iteration, and when registration and calibration are completed. The camera position is fixed using PCL for all screenshots. (b) Error distribution of returned hand–eye relations. Each box shows the first quartile to the third quartile of 100 values, with caps showing minimum and maximum error.

of multiview point clouds under the robot base frame by iterative refinement of hand–eye relation. Experimenting with a real eye-in-hand system proves the proposed method feasible and effective. We hope our work can make on-site calibration flexible and easy when necessary.

In the future, we plan to further investigate the impact of sampling density on calibration performance and jointly consider the robot absolute positioning error. We also intend to test our method on other 3-D sensor and quasi-hand–eye configurations, such as Lidar-IMU system, if possible.

References

[1] H. Zhang et al., "A practical robotic grasping method by using 6-D pose estimation with protective correction," *IEEE Trans. Ind. Electron.*, vol. 69, no. 4, pp. 3876–3886, Apr. 2022, doi: 10.1109/TIE.2021.3075836.

[2] W. Lin, A. Anwar, Z. Li, M. Tong, J. Qiu, and H. Gao, "Recognition and pose estimation of auto parts for an autonomous spray painting robot," *IEEE Trans. Ind. Informat.*, vol. 15, no. 3, pp. 1709–1719, Mar. 2019, doi: 10.1109/TII.2018.2882446.

[3] P. Zhou, R. Peng, M. Xu, V. Wu, and D. Navarro-Alarcon, "Path planning with automatic seam extraction over point cloud models for robotic arc welding," *IEEE Robot. Automat. Lett.*, vol. 6, no. 3, pp. 5002–5009, Jul. 2021, doi: 10.1109/LRA.2021.3070828.

[4] Y. Shiu and S. Ahmad, "Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form AX=XB," *IEEE Trans. Robot. Automat.*, vol. 5, no. 1, pp. 16–29, Feb. 1989, doi: 10.1109/70.88014.

[5] R. Tsai and R. Lenz, "A new technique for fully autonomous and efficient 3D robotics hand/eye calibration," *IEEE Trans. Robot. Automat.*, vol. 5, no. 3, pp. 345–358, Jun. 1989, doi: 10.1109/70.34770.

[6] J. C. Chou and M. Kamel, "Finding the position and orientation of a sensor on a robot manipulator using quaternions," *Int. J. Robot. Res.*, vol. 10, no. 3, pp. 240–254, 1991.

[7] F. Park and B. Martin, "Robot sensor calibration: Solving AX=XB on the Euclidean group," *IEEE Trans. Robot. Automat.*, vol. 10, no. 5, pp. 717–721, Oct. 1994, doi: 10.1109/70.326576.

[8] K. Daniilidis, "Hand-eye calibration using dual quaternions," *Int. J. Robot. Res.*, vol. 18, no. 3, pp. 286–298, 1999.

[9] N. Andreff, R. Horaud, and B. Espiau, "On-line hand-eye calibration," in *Proc. IEEE 2nd Int. Conf. 3-D Digit. Imag. Model.*, 1999, pp. 430–436.

[10] H. Zuang and Y. C. Shiu, "A noise-tolerant algorithm for robotic hand-eye calibration with or without sensor orientation measurement," *IEEE Trans. Syst., Man, Cybern.*, vol. 23, no. 4, pp. 1168–1175, Jul./Aug. 1993, doi: 10.1109/21.247898.

[11] R. Horaud and F. Dornaika, "Hand-eye calibration," *Int. J. Robot. Res.*, vol. 14, no. 3, pp. 195–210, 1995.

[12] K. H. Strobl and G. Hirzinger, "Optimal hand-eye calibration," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2006, pp. 4647–4653, doi: 10.1109/IROS.2006.282250.

[13] Z. Zhao, "Hand-eye calibration using convex optimization," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2011, pp. 2947–2952, doi: 10.1109/ICRA.2011.5979569.

[14] X. Wang, J. Huang, and H. Song, "Simultaneous robot–world and hand–eye calibration based on a pair of dual equations," *Measurement*, vol. 181, 2021, Art. no. 109623.

[15] E. Pedrosa, M. Oliveira, N. Lau, and V. Santos, "A general approach to hand eye calibration through the optimization of atomic transformations," *IEEE Trans. Robot.*, vol. 37, no. 5, pp. 1619–1633, Oct. 2021, doi: 10.1109/TRO.2021.3062306.

[16] J. Fu, Y. Ding, T. Huang, H. Liu, and X. Liu, "Hand–eye calibration method based on three-dimensional visual measurement in robotic high-precision machining," *Int. J. Adv. Manuf. Technol.*, vol. 119, no. 5, pp. 3845–3856, 2022.

[17] J. Fu, Y. Ding, T. Huang, and X. Liu, "Hand-eye calibration method with a three-dimensional-vision sensor considering the rotation parameters of the robot pose," *Int. J. Adv. Robot. Syst.*, vol. 17, no. 6, Nov. 2020, doi: 10.1177/1729881420977296.

[18] X. Liu et al., "Fast eye-in-hand 3-D scanner-robot calibration for low stitching errors," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8422–8432, Sep. 2021, doi: 10.1109/TIE.2020.3009568.

[19] S. Kahn, D. Haumann, and V. Willert, "Hand-eye calibration with a depth camera: 2D or 3D?," in *Proc. IEEE Int. Conf. Comput. Vis. Theory Appl.*, 2014, vol. 3, pp. 481–489.

[20] H. Xie, W. Li, and H. Liu, "General geometry calibration using arbitrary free-form surface in a vision-based robot system," *IEEE Trans. Ind. Electron.*, vol. 69, no. 6, pp. 5994–6003, Jun. 2022, doi: 10.1109/TIE.2021.3090716.

[21] B. Kaiser, R. A. Tauro, and H. Worn, "Extrinsic calibration of a robot mounted 3D imaging sensor," *Int. J. Intell. Syst. Technol. Appl.*, vol. 5, no. 3/4, pp. 374–379, 2008.

[22] R. S. Andersen, O. Madsen, and T. B. Moeslund, "Hand-eye calibration of depth cameras based on planar surfaces," in *Proc. Int. Workshop Intell. Robot Assistants: Part IRAS 2014*, 2014.

[23] G. Xu and Y. Yan, "A scene feature based eye-in-hand calibration method for industrial robot," in *Mechatronics and Machine Vision in Practice*, vol. 4. Berlin, Germany: Springer, 2021, pp. 179–191.

[24] W. Peng, Y. Wang, H. Zhang, Q. Zhu, Z. Miao, and M. Feng, "Stochastic joint alignment of multiple point clouds for profiled blades 3-D reconstruction," *IEEE Trans. Ind. Electron.*, vol. 69, no. 2, pp. 1682–1693, Feb. 2022, doi: 10.1109/TIE.2021.3059538.

[25] Z. Wang, J. Fan, F. Jing, S. Deng, M. Zheng, and M. Tan, "An efficient calibration method of line structured light vision sensor in robotic eye-in-hand system," *IEEE Sensors J.*, vol. 20, no. 11, pp. 6200–6208, Jun. 2020.

[26] R. Hartley, J. Trumpf, Y. Dai, and H. Li, "Rotation averaging," *Int. J. Comput. Vis.*, vol. 103, no. 3, pp. 267–305, 2013.

[27] P. Besl and N. D. McKay, "A method for registration of 3-D shapes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 14, no. 2, pp. 239–256, Feb. 1992, doi: 10.1109/34.121791.

[28] J. Wu, Y. Sun, M. Wang, and M. Liu, "Hand-eye calibration: 4-D procrustes analysis approach," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 2966–2981, Jun. 2020, doi: 10.1109/TIM.2019.2930710.

[29] J. Xu, J. L. Hoo, S. Dritsas, and J. G. Fernandez, "Hand-eye calibration for 2D laser profile scanners using straight edges of common objects," *Robot. Comput.-Integr. Manuf.*, vol. 73, 2022, Art. no. 102221.

[30] R. B. Rusu, N. Blodow, and M. Beetz, "Fast point feature histograms (FPFH) for 3D registration," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2009, pp. 3212–3217, doi: 10.1109/ROBOT.2009.5152473.

[31] J. Tremblay, S. Tyree, T. Mosier, and S. Birchfield, "Indirect object-to-robot pose estimation from an external monocular RGB camera," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 4227–4234, doi: 10.1109/IROS45743.2020.9341163.

[32] Z. Wang et al., "A pose estimation system based on deep neural network and ICP registration for robotic spray painting application," *Int. J. Adv. Manuf. Technol.*, vol. 104, no. 1, pp. 285–299, 2019.

Shiyu Xing received the B.S. degree in automotive engineering from Nanchang University, Jiangxi, China, in 2016, and the M.S. degree in mechanical engineering from Washington University in St. Louis, St. Louis, MO, USA, in 2018. He is currently working toward the Ph.D. degree in computer application technology with the Institute of Automation, Chinese Academy of Sciences, Beijing, China.

He was a Car2X Engineer with Mercedes Benz R&D, Beijing, China, from 2018 to 2020. He is currently with the School of Artificial Intelligence, University of Chinese Academy of Sciences. His research interests include robotics, state estimation, and calibration.

Fengshui Jing received the B.S. degree in mining engineering from Huainan Mining Institute, Anhui, China, in 1991, the M.S. degree in safety technology and engineering from Shandong Mining Institute, Shandong, China, in 1994, and the Ph.D. degree in control theory and control engineering from the Institute of Automation, Chinese Academy of Sciences (IACAS), Beijing, China, in 2002.

He is currently a Professor with the State Key Laboratory of Management and Control for Complex Systems, IACAS. His research interests include robotics, computer vision, and manufacturing systems.

Min Tan received the B.S. degree from Tsing Hua University, Beijing, China, in 1986, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences (IACAS), Beijing, in 1990, both in control science and engineering.

He is currently a Professor with the State Key Laboratory of Management and Control for Complex Systems, IACAS. His research interests include robotics and intelligent control system.